



# Department Of Computer Science, CUI Lahore Campus

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CSC102 - Discrete Structures

# Lecture Outline

- Predicate Logic
  - Predicate
  - Quantifier
  - Translation of Quantified Statements

# Predicate Logic

- Proposition, YES or NO?

- $3 + 2 = 5$  Yes

- $X + 2 = 5$  No

- $X + 4 = 5$  for some  $X$  in  $\{1, 2, 3\}$  Yes

- Computer  $X$  is under attack by an intruder No

# Why Predicate Logic?

- **Propositional Logic is not expressive enough**
  - It cannot adequately express the meaning of statements in mathematics and in natural language

## Example 1:

**“Every computer connected to the university network is functioning properly.”**

- No rules of propositional logic allow us to conclude the truth of the statement.

# Why Predicate Logic?

## Example 2:

- “There is a computer on the university network that is under attack by an intruder.”

**Predicate Logic is more expressive  
and powerful**

# Propositional Functions(Example)

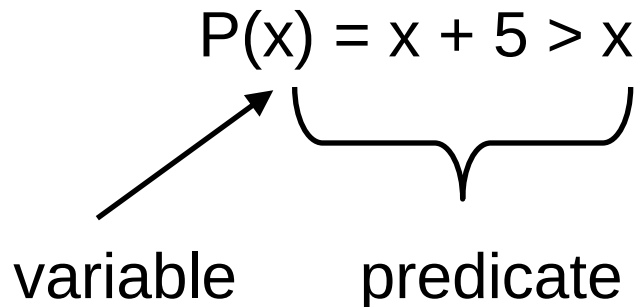
- “x is greater than 3” or  $(x > 3)$ 
  - The variable x: subject of the statement
  - “is greater than 3”: predicate
  - $P(x)$ : propositional function P at x
- Let  $P(x) = x > 3$ 
  - $P(x)$  has no truth values (x is not given a value)
  - $P(10)$  is true: The proposition  $10 > 3$  is true.
  - $P(1)$  is false: The proposition  $1 > 3$  is false.
  - $P(x)$  will create a proposition when given a value

## Propositional Functions(Example)

- Let  $A(x)$  = “Computer  $x$  is under attack by an intruder.”
- Suppose computers on campus, only CS2 and MATH1 are currently under attack by intruders.
- What are truth values of  $A(\text{CS1})$ ,  $A(\text{CS2})$ , and  $A(\text{MATH1})$ ?
- The statement  $A(\text{CS1})$  by setting  $x = \text{CS1}$  in the statement “Computer  $x$  is under attack by an intruder.”
- CS1 is not on the list of computers currently under attack,  $A(\text{CS1})$  is false.
- CS2 and MATH1 are on the list of computers under attack,  $A(\text{CS2})$  and  $A(\text{MATH1})$  are true.

# Propositional Functions

- Functions with multiple variables:
  - $P(x,y) = x + y == 0$ 
    - $P(1,2)$  is false,  $P(1,-1)$  is true
  - $P(x,y,z) = x + y == z$ 
    - $P(3,4,5)$  is false,  $P(1,2,3)$  is true
  - $P(x_1, x_2, x_3 \dots x_n) = \dots$
- Anatomy of a propositional function





# Predicates

- A predicate is a declarative statement with at least one variable (i.e. unknown value).
- A predicate, or propositional function, is a function that takes some variable(s) as arguments and returns True or False.

# Predicates

- Suppose  $Q(x,y) = "x > y"$

Proposition, YES or NO?

$Q(x,y)$

No

$Q(3,4)$

Yes

$Q(x,9)$

No

Predicate, YES or NO?

$Q(x,y)$

Yes

$Q(3,4)$

No

$Q(x,9)$

Yes

# Quantification

- Quantification expresses the extent to which a predicate is true over a range of elements.
- In English, the words *all*, *some*, *many*, *none*, and *few* are used in quantifications.
- The area of logic that deals with predicates and quantifiers is called the **predicate calculus**.

## Types of Quantifiers

- A quantifier is “an operator that limits the variables of a proposition”.
- Two types:
  - Universal
  - Existential

# Universal Quantifiers

- Represented by an upside-down A:  $\forall$ 
  - It means “for all”
  - Let  $P(x) = x+1 > x$
- We can state the following:
  - $\forall x P(x)$
  - English translation: “for all values of  $x$ ,  $P(x)$  is true”
  - English translation: “for all values of  $x$ ,  $x+1 > x$  is true”

Besides “for all”, universal quantification can be expressed in many other ways: “for every”, “all of”, “for each”, “given any”, “for arbitrary”, “for each” and “for any”

# Universal Quantifiers

- You need to specify the **universe of quantification**!
  - What values  $x$  can represent
  - Called the “domain of discourse” or “universe of discourse”
  - Or just “domain” or “universe”
- The meaning of the universal quantification of  $P(x)$  changes when we change the domain. The domain must always be specified when a universal quantifier is used; without it, the universal quantification of a statement is not defined.

# Universal Quantifiers

- Let the universe of discourse be the real numbers.
- Let  $P(x) = x/2 < x$ 
  - Not true for the negative numbers!
  - Thus,  $\forall x P(x)$  is false, When the domain is all the real numbers
- In order to prove that a universal quantification is true, it must be shown for **ALL** cases
- In order to prove that a universal quantification is false, it must be shown to be false for **only ONE** case

# Universal Quantifiers

- Let  $P(x)$  is “ $x^2 > 0$ .” To show that the statement  $\forall x P(x)$  is false where the universe of discourse consists of all integers, we give a counterexample.
- $x = 0$  is a counterexample because  $x^2 = 0$  when  $x = 0$ , so that  $x^2$  is not greater than 0 when  $x = 0$ .



# Universal Quantification

- Given some propositional function  $P(x)$  And values in the universe  $x_1 \dots x_n$
- The universal quantification  $\forall x P(x)$  implies:
- $P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$

## Question

- What is the truth value of  $\forall x P(x)$ , where  $P(x)$  is the statement  $x^2 < 10$  and the domain consists of the positive integers not exceeding 4?

## Solution:

- The statement  $\forall x P(x)$  is the same as the conjunction  $P(1) \wedge P(2) \wedge P(3) \wedge P(4)$ ,
- Because  $P(4) \equiv 4^2 < 10$ , is **false**, it follows that  $\forall x P(x)$  is **false**.

# Existential Quantification

- Represented by an backwards E:  $\exists$ 
  - It means “there exists”, “there is”, “for some”, etc.
  - Let  $P(x) = x+1 > x$
- We can state the following:
  - $\exists x P(x)$
  - English translation: “there exists (a value of)  $x$  such that  $P(x)$  is true”
  - English translation: “for at least one value of  $x$ ,  $x+1 > x$  is true”
  - English translation: “for some  $x$ ,  $P(x)$ ”

# Existential Quantification

- Let  $P(x) = x+1 > x$ 
  - There is a numerical value for which  $x+1 > x$
  - In fact, it's true for all of the values of  $x$ . Thus,  $\exists x$   $P(x)$  is true
- In order to show an existential quantification is **true**, you only have to **find ONE value**
- In order to show an existential quantification is **false**, you have to show **it's false for ALL values**

# Existential Quantification

- **Example:** Let  $P(x)$  denote the statement “ $x > 3$ .” What is the truth value of the quantification  $\exists x P(x)$ , where the domain consists of all real numbers?
- **Solution:** Because “ $x > 3$ ” is sometimes true—for instance, when  $x = 4$  the existential quantification of  $P(x)$ , which is  $\exists x P(x)$ , is true.

# Existential Quantification

- **Example:** Let  $Q(x)$  denote the statement " $x = x + 1$ ." What is the truth value of the quantification  $\exists x Q(x)$ , where the domain consists of all real numbers?
- **Solution:** Because  $Q(x)$  is false for every real number  $x$ , the existential quantification of  $Q(x)$ , which is  $\exists x Q(x)$ , is false.

# Existential Quantification

- Given some propositional function  $P(x)$  And values in the universe  $x_1 \dots x_n$
- The existential quantification  $\exists x P(x)$  implies:
- $P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$

# Summary

| Statement        | When True?                                | When False?                                |
|------------------|-------------------------------------------|--------------------------------------------|
| $\forall x P(x)$ | $P(x)$ is true for every $x$ .            | There is an $x$ for which $P(x)$ is false. |
| $\exists x P(x)$ | There is an $x$ for which $P(x)$ is true. | $P(x)$ is false for every $x$ .            |



## Quantifiers with Restricted Domain

- An abbreviated notation is often used to restrict the domain of a quantifier.
- In this notation, a condition a variable must satisfy is included after the quantifier.
- $\forall x < 0 (x^2 > 0)$  where domain is real numbers

## Quantifiers with Restricted Domain

- $\forall x < 0 (x^2 > 0) \equiv \forall x ((x < 0) \rightarrow (x^2 > 0))$
- The restriction of a universal quantification is the same as the universal quantification of a conditional statement.

# Quantifiers with Restricted Domain

- $\forall y \neq 0 (y^3 \neq 0) \equiv \forall y (y \neq 0 \rightarrow y^3 \neq 0)$

# Quantifiers with Restricted Domain

- $\exists z > 0 (z^2 = 2) \equiv \exists z (z > 0 \wedge z^2 = 2)$
- The restriction of an existential quantification is the same as the existential quantification of a conjunction.

# Precedence of Quantifiers

- The quantifiers  $\forall$  and  $\exists$  have higher precedence than all logical operators from propositional calculus.
- e.g  $\forall x P(x) \vee Q(x)$  is the disjunction of  $\forall x P(x)$  and  $Q(x)$ .

## Binding Variables

- When a quantifier is used on a variable  $x$ , we say that this occurrence of the variable is **bound**.
- An occurrence of a variable that is not bound by a quantifier or not set equal to a particular value is said to be **free**.
- The part of a logical expression to which a quantifier is applied is called the **scope** of the quantifier.
- All the variables that occur in a logical expression must be bound or set equal to a particular value to turn into a proposition.

# Binding Variables

- Examples:
- $P(x)$              $x$  is free
- $P(5)$              $x$  is bound to 5
- $\forall x P(x)$          $x$  is bound by quantifier

## Binding Variables

- $\exists x (P(x) \wedge Q(x)) \vee (\forall x R(x))$ 
  - All variables are bound.
- The scope of the first quantifier,  $\exists x$ , is the expression  $P(x) \wedge Q(x)$  because  $\exists x$  is applied only to  $P(x) \wedge Q(x)$ , and not to the rest of the statement.
- Similarly, the scope of the second quantifier,  $\forall x$ , is the expression  $R(x)$ .
- That is, the existential quantifier binds the variable  $x$  in  $P(x) \wedge Q(x)$  and the universal quantifier  $\forall x$  binds the variable  $x$  in  $R(x)$ .



## Binding Variables

- $\exists x (x + y = 1)$ 
  - $x$  is bound by  $\exists x$  and  $y$  is free; thus not a proposition
- $(\exists x P(x)) \vee Q(x)$ 
  - The  $x$  in  $Q(x)$  is not bound; thus not a proposition
- $(\exists x P(x)) \vee (\forall x Q(x))$ 
  - Both  $x$  values are bound; thus it is a proposition
- $\exists x (P(x) \wedge Q(x)) \vee (\forall y R(y))$ 
  - All variables are bound; thus it is a proposition
- $(\exists x P(x) \wedge Q(y)) \vee (\forall y R(y))$ 
  - The  $y$  in  $Q(y)$  is not bound; thus not a proposition

## A note on quantifiers

- Recall that  $P(x)$  is a propositional function
  - Let  $P(x)$  be “ $x == 0$ ”
- Recall that a proposition is a statement that is either true or false
  - $P(x)$  is not a proposition
- There are two ways to make a propositional function into a proposition:
  - Supply it with a value
    - For example,  $P(5)$  is false,  $P(0)$  is true
  - Provide a quantification
    - For example,  $\forall x P(x)$  is false and  $\exists x P(x)$  is true
    - Let the universe of discourse be the real numbers

# Translating From English to Logical Expressions

- Express the statement “Every student in this class has studied calculus” using predicates and quantifiers.

- **Solution:**

- Assume domain is students in the class

“For every student in this class, that student has studied calculus.”

“For every student  $x$  in this class,  $x$  has studied calculus.”

$C(x)$  = “ $x$  has studied calculus.”

$$\forall x C(x)$$

## Translating From English to Logical Expressions

- Express the statement “Every student in this class has studied calculus” using predicates and quantifiers.
- *Let*  $C(x)$  = “ $x$  has studied calculus.”  
 $S(x)$  = “person  $x$  is student in this class.”  
The domain for  $x$  consists of all people.
- “For every person  $x$ , if person  $x$  is a student in this class then  $x$  has studied calculus.”
- The statement can be expressed as  $\forall x(S(x) \rightarrow C(x))$ .

# Negating Quantified Expressions

- Consider the statement  
“Every student in this class has studied calculus.”
- This statement is a universal quantification, namely,  $\forall x C(x)$ ,
  - $C(x)$  is the statement “ $x$  has studied calculus”
  - Domain consists of the students in the class.
- The negation of this statement is
  - “It is not the case that every student in this class has studied calculus.”
  - This is equivalent to “There is a student in this class who has not studied calculus.”
- This is simply the existential quantification of the negation of the original propositional function, namely,  $\exists x \neg C(x)$ .

# Negating Quantified Expressions

| Statement        | When True?                                | When False?                                |
|------------------|-------------------------------------------|--------------------------------------------|
| $\forall x P(x)$ | $P(x)$ is true for every $x$ .            | There is an $x$ for which $P(x)$ is false. |
| $\exists x P(x)$ | There is an $x$ for which $P(x)$ is true. | $P(x)$ is false for every $x$ .            |

| Negation              | Equivalent Statement  | When is Negation True?                     | When False?                               |
|-----------------------|-----------------------|--------------------------------------------|-------------------------------------------|
| $\neg \exists x P(x)$ | $\forall x \neg P(x)$ | For every $x$ , $P(x)$ is false.           | There is an $x$ for which $P(x)$ is true. |
| $\neg \forall x P(x)$ | $\exists x \neg P(x)$ | There is an $x$ for which $P(x)$ is false. | $P(x)$ is true for every $x$ .            |

# Negating Quantified Expressions

- **Example:**

What is the negation of the statement “There is an honest politician”?

**Solution:**

- Let  $H(x)$  denote “ $x$  is honest.”
- The statement is represented by  $\exists xH(x)$ , where the domain consists of all politicians.
- The negation of this statement is  $\neg\exists xH(x)$ , which is equivalent to  $\forall x\neg H(x)$ .
- This negation can be expressed as “Every politician is dishonest.” or “Not all politicians are honest.”

# Negating Quantified Expressions

## Example:

What is the negation of the statement “All Americans eat cheeseburgers”?

## Solution:

- $C(x)$  denote “ $x$  eats cheeseburgers.”
- The statement is represented by  $\forall x C(x)$ , where the domain consists of all Americans.
- The negation of this statement is  $\neg \forall x C(x)$ , which is equivalent to  $\exists x \neg C(x)$ .
- This negation can be expressed as “Some American does not eat cheeseburgers” and “There is an American who does not eat cheeseburgers.”



# Negating Quantified Expressions

- What are the negations of the statements  $\forall x(x^2 > x)$  and  $\exists x(x^2 = x)$ ?

$$\forall x(x^2 > x)$$

$$\equiv \neg \forall x(x^2 > x)$$

$$\equiv \exists x \neg(x^2 > x)$$

$$\equiv \exists x(x^2 \leq x)$$

$$\exists x(x^2 = x)$$

$$\equiv \neg \exists x(x^2 = x)$$

$$\equiv \forall x \neg(x^2 = x)$$

$$\equiv \forall x(x^2 \neq x)$$

# De Morgan's Laws for Quantifiers

- $\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n)$
- $\exists x P(x) \equiv P(x_1) \vee P(x_2) \vee \dots \vee P(x_n)$
- $\neg \exists x P(x) \equiv \neg(P(x_1) \vee P(x_2) \vee \dots \vee P(x_n))$   
 $\equiv \neg P(x_1) \wedge \neg P(x_2) \wedge \dots \wedge \neg(P(x_n))$   
 $\equiv \forall x \neg P(x)$
- $\neg \forall x P(x) \equiv \neg(P(x_1) \wedge P(x_2) \wedge \dots \wedge P(x_n))$   
 $\equiv \neg P(x_1) \vee \neg P(x_2) \vee \dots \vee \neg(P(x_n))$   
 $\equiv \exists x \neg P(x)$

## Translating From English to Logical Expressions

- Let  $R(x)$  = “ $x$  can speak Russian”  
 $C(x)$  = “ $x$  knows the computer language C++.”

Express each of these sentences in terms of  $R(x)$ ,  $C(x)$ , quantifiers, and logical connectives.

The domain for quantifiers consists of all students at your school.

- There is a student at your school who can speak Russian and who knows C++.

$$\exists x(R(x) \wedge C(x))$$

## Translating From English to Logical Expressions

- Let  $R(x)$  = “ $x$  can speak Russian”  
 $C(x)$  = “ $x$  knows the computer language C++.”
- There is a student at your school who can speak Russian but who doesn't know C++.

$$\exists x(R(x) \wedge \neg C(x))$$

## Translating From English to Logical Expressions

- Let  $R(x)$  = “ $x$  can speak Russian”  
 $C(x)$  = “ $x$  knows the computer language C++.”
- Every student at your school either can speak Russian or knows C++.

$$\forall x(R(x) \vee C(x))$$

## Translating From English to Logical Expressions

- Let  $R(x)$  = “ $x$  can speak Russian”  
 $C(x)$  = “ $x$  knows the computer language C++.”
- No student at your school can speak Russian or knows C++.

$$\forall x \neg (R(x) \vee C(x)) / \neg \exists x (R(x) \vee C(x))$$

# The Four Aristotelian Forms

1. All A's are B's
  2. Some A's are B's
  3. No A's are B's
  4. Some A's are not B's
- These are four of the most common quantificational sentences used in quantificational reasoning.

# The First Aristotelian Form

- The Form: *All A's are B's*
- Example: All comedian are funny.
  - Rephrase: For every  $x$ , if  $x$  is a comedian then  $x$  is funny
  - Translation:  $\forall x (\text{Comedian}(x) \rightarrow \text{Funny}(x))$
  - This translation has the form:  $\forall x (A(x) \rightarrow B(x))$
- General Fact
  - All A's are B's translates as  $\forall x (A(x) \rightarrow B(x))$



## The Second Aristotelian Form

- The Form: *Some A's are B's*
- Example: Some comedian are funny
  - Rephrase: Some thing  $x$  is both comedian and funny
  - Translation:  $\exists x (\text{Comedian}(x) \wedge \text{Funny}(x))$
  - This translation has the form:  $\exists x (A(x) \wedge B(x))$
- General Fact
  - Some A's are B's translates as  $\exists x (A(x) \wedge B(x))$

# The Third Aristotelian Form

- The Form: *No A's are B's*
- Example: No students are failed
  - Rephrase: For every  $x$ , if  $x$  is a student then  $x$  is not failed
  - Translation:  $\forall x (\text{Student}(x) \rightarrow \neg \text{Failed}(x))$
  - This translation has the form:  $\forall x (A(x) \rightarrow \neg B(x))$
- General Fact
  - No A's are B's translates as  $\forall x (A(x) \rightarrow \neg B(x))$

# The Fourth Aristotelian Form

- The Form: *Some A's are not B's*
- Example: Some excuses are not believable
  - Rephrase: For some  $x$ ,  $x$  is an excuse and  $x$  is not believable
  - Translation:  $\exists x (\text{Excuse}(x) \wedge \neg \text{Believable}(x))$
  - This translation has the form:  $\exists x (A(x) \wedge \neg B(x))$
- General Fact
  - Some A's are not B's translates as  $\exists x (A(x) \wedge \neg B(x))$

# Summary

- The Aristotelian Forms and Their Translations
  - All A's are B's  $\forall x (A(x) \rightarrow B(x))$
  - Some A's are B's  $\exists x (A(x) \wedge B(x))$
  - No A's are B's  $\forall x (A(x) \rightarrow \neg B(x))$
  - Some A's are not B's  $\exists x (A(x) \wedge \neg B(x))$

## Predicates - Examples

$L(x)$  = “x is a lion.”

$F(x)$  = “x is fierce.”

$C(x)$  = “x drinks coffee.”

Assuming that the domain consists of all creatures.

- All lions are fierce.

$$\forall x (L(x) \rightarrow F(x))$$

- Some lions don't drink coffee.

$$\exists x (L(x) \wedge \neg C(x))$$

- Some fierce creatures don't drink coffee.

$$\exists x (F(x) \wedge \neg C(x))$$

## Predicates - Examples

$B(x)$  = “x is a hummingbird.”

$L(x)$  = “x is a large bird.”

$H(x)$  = “x lives on honey.”

$R(x)$  = “x is richly colored.”

Assuming that the domain consists of all birds.

- All hummingbirds are richly colored.

$$\forall x (B(x) \rightarrow R(x))$$

- No large birds live on honey.

$$\forall x (L(x) \rightarrow \neg H(x))$$

- Birds that do not live on honey are dully colored.

$$\forall x (\neg H(x) \rightarrow \neg R(x))$$

- Hummingbirds are small.

$$\forall x (B(x) \rightarrow \neg L(x))$$

## Example

- Express each of these statements using quantifiers. Then form the negation of the statement, so that no negation is to the left of a quantifier. Next, express the negation in simple English as well.
  - a) Some old dogs can learn new tricks.
  - b) No rabbit knows calculus.
  - c) Every bird can fly.
  - d) There is no dog that can talk.
  - e) There is no one in this class who knows French and Russian.

# Chapter Reading

- **Chapter 1**, Kenneth H. Rosen, Discrete Mathematics and Its Applications, Section 1.4



# Chapter Exercise (For Practice)

- Question # 1, 2, 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 17, 18, 33, 34, 61(a, b, c), 62( a, b, c), 63(a, b, c, d)